

Novel risk assessment model of food quality and safety considering physical-chemical and pollutant indexes based on coefficient of variance integrating entropy weight

A. Reviewer^a, B. Author^{b,*}

Abstract. Food quality and safety risk assessment combines several physical-chemical and pollutant indexes into one score. We weight each index by its coefficient of variance and entropy, then rank samples by the combined risk value.

1 Introduction

Risk assessment of food depends on indexes that vary in scale and spread. A weight that follows the data is easier to defend than one set by hand. The model below uses the variation of each index to set its weight.

Samples were scored on each index, scaled to the same range, and combined. The result is one number per sample that regulators can compare across lots and sites.

2 Methods

Each index was normalized and weighted by its share of total variation. Entropy of the normalized index adjusted the weight for indexes that carried little information about the samples.

Sampling followed the plan used for routine lot release. Results were checked against the limits in force at the time of testing.

3 Results

Weighted risk scores separated the high-risk lots from the rest in every case. The ranking changed little when any one index was removed from the model.